

9 Hierarchical Clustering

MA2003B Application of Multivariate Methods
in Data Science

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1. **Hierarchical Clustering**: These methods create a **dendrogram** (also known as a **tree diagram**) representing the taxonomic structure of our data. It is specially useful for small datasets and when the number of clusters is unknown.
2. **Non-Hierarchical Clustering**: These methods, such as K-means and DBSCAN, partition the data into a predefined number of clusters or based on density. They are generally more scalable for larger datasets.

Understanding Dendrograms

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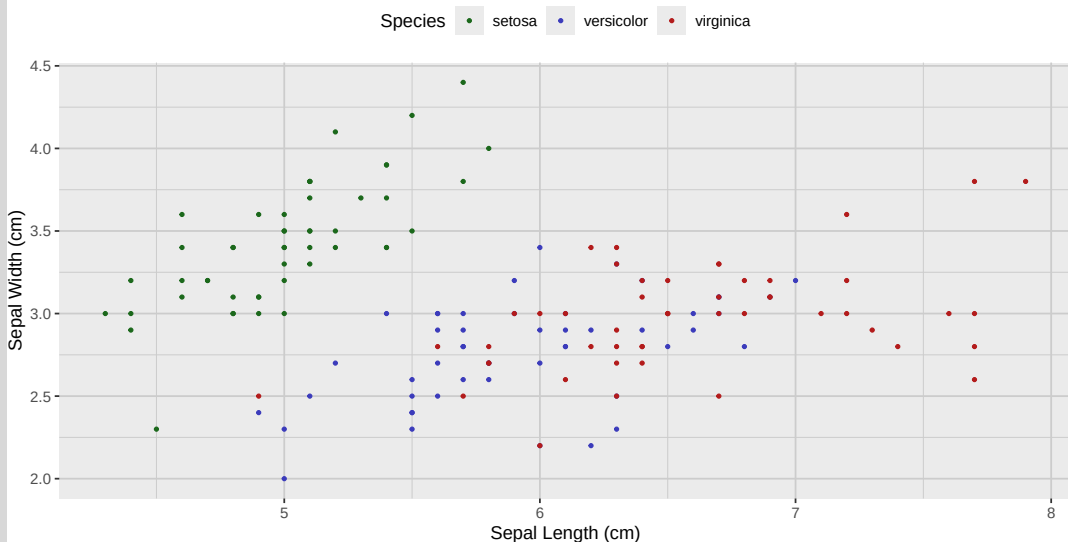
The vertical axis represents the distance or dissimilarity between clusters, while the horizontal axis represents the individual data points or clusters.

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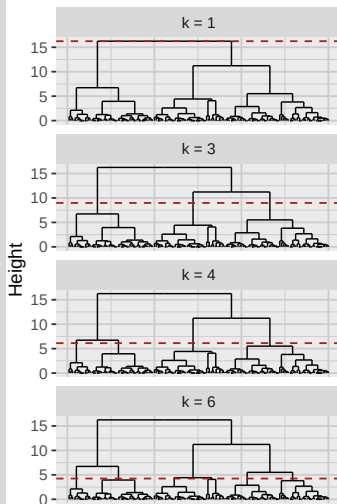
As an illustration, let's consider the scatter plot of the Iris dataset (considering only the variables "Sepal.Length" and "Sepal.Width") where each point is colored according to the species of the flower:

Iris Dataset: Sepal Length vs Sepal Width

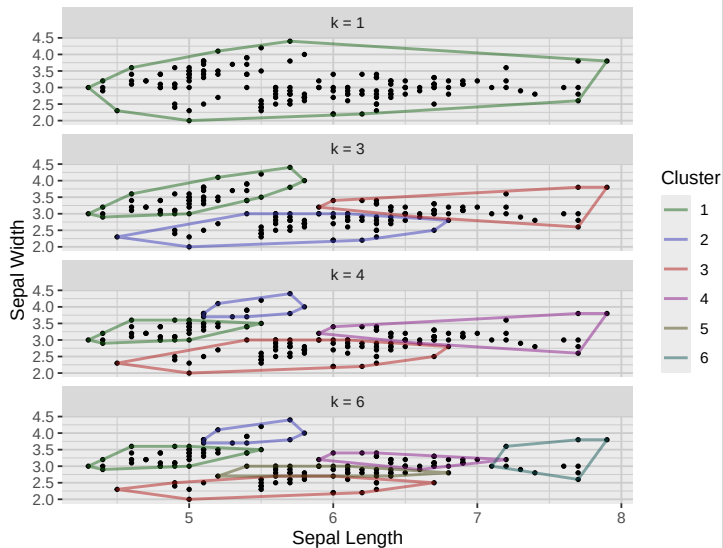


We will now consider a dendrogram associated to this dataset and show how cutting it at different heights leads to different numbers of clusters:

Dendrogram cuts



Clusters



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- **Agglomerative Clustering:** This is a bottom-up approach where each data point starts as its own cluster, and pairs of clusters are merged iteratively based on their similarity until a single cluster is formed or a stopping criterion is met.
- **Divisive Clustering:** This is a top-down approach where all data points start in a single cluster, and the cluster is recursively split into smaller clusters based on dissimilarity until each data point is its own cluster or a stopping criterion is met.

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Common distance metrics include:

- **Euclidean Distance:** This is the most common distance metric, which calculates the straight-line distance between two points in a multi-dimensional space.

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- **Mahalanobis Distance:** This metric takes into account the correlations between variables and is useful when dealing with multivariate data.
- **Canberra Distance:** This metric is sensitive to small changes in the data and is useful when dealing with sparse data. The formula for the Canberra distance between two points p and q is given by:

$$d(p, q) = \sum_{i=1}^n \frac{|p_i - q_i|}{|p_i| + |q_i|}$$

where n is the number of dimensions, and p_i and q_i are the coordinates of points p and q respectively.

- **Manhattan Distance:** Also known as the L^1 norm or taxicab distance, this metric calculates the distance between two points by summing the absolute differences of their coordinates. It is particularly useful when dealing with high-dimensional data or when the data has a grid-like structure. The formula for the Manhattan distance between two points p and q is given by:

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- **Minkowski Distance:** This is a generalization of the Euclidean and Manhattan distances, which allows for different values of the parameter p to control the distance calculation. The formula for the Minkowski distance between two points p and q is given by:

$$d(p, q) = \left(\sum_{i=1}^n |p_i - q_i|^p \right)^{1/p}$$

When $p = 1$, it corresponds to the Manhattan distance, and when $p = 2$, it corresponds to the Euclidean distance.

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- **Single Linkage:** This criterion defines the distance between two clusters as the minimum distance between any pair of points from each cluster. It tends to create long, chain-like clusters and is sensitive to noise and outliers.
- **Complete Linkage:** This criterion defines the distance between two clusters as the maximum distance between any pair of points from each cluster. It tends to create compact, spherical clusters and is less sensitive to noise and outliers compared to single linkage.

- **Average Linkage:** This criterion defines the distance between two clusters as the average distance between all pairs of points from each cluster. It provides a balance between single and complete linkage and is less sensitive to noise and outliers.

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- **Ward's Linkage:** This criterion minimizes the total within-cluster variance by merging clusters that result in the smallest increase in variance. It tends to create compact, spherical clusters and is less sensitive to noise and outliers compared to other linkage criteria.

Once we have chosen the method, distance metric, and linkage criterion, we can perform hierarchical clustering using the `hclust` function in R.

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As was shown in the previous example, we can then visualize the results using a dendrogram and cut the dendrogram at a specific height to determine the number of clusters.